

PREOBRAZHENIYE, Russian Federation

WMO: 319890

Lat: 42.9005N

Lon: 133.8921E

Elev: 43

StdP: 100.81

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-14.7	-29.8	0.2	-13.9	-28.0	0.3	-12.6	10.2	-6.3	8.2	-10.0	3.3	0	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	24.9	21.1	23.7	20.7	22.6	20.1	22.3	23.8	21.5	23.0	20.7	22.2	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	16.5	23.2	20.9	15.7	22.5	20.1	14.8	21.8	66.0	23.9	63.0	23.3	60.2	22.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.5	6.0	-19.3	27.7	2.2	0.9	-20.9	28.4	-22.2	28.9	-23.5	29.4	-25.1	30.0		
			DB	WB	-20.3	23.5	2.0	1.0	-21.7	24.2	-22.9	24.8	-24.0	25.3	-25.4	26.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-7.3	-5.1	-0.2	5.0	9.5	12.8	17.8	20.1	16.3	9.6	1.4	-5.5
	DBStd	9.73	3.68	3.75	3.18	2.61	2.86	2.36	2.55	2.15	2.45	3.34	4.66	4.41
	HDD10.0	2261	538	422	317	151	45	3	0	0	0	46	259	480
	HDD18.3	4498	796	655	575	400	273	165	41	8	68	269	509	738
	CDD10.0	897	0	0	0	1	30	88	241	312	189	35	1	0
	CDD18.3	93	0	0	0	0	0	1	23	62	8	0	0	0
	CDH23.3	142	0	0	0	0	0	1	25	110	6	0	0	0
	CDH26.7	6	0	0	0	0	0	0	1	6	0	0	0	0
Wind	WSAvg	2.6	2.9	2.6	2.5	2.6	2.5	2.4	2.2	2.5	2.9	2.7	2.8	2.9
Precipitation	PrecAvg	781	12	20	30	44	80	75	141	135	112	59	46	28
	PrecMax	1131	33	71	65	133	176	158	426	282	351	127	154	137
	PrecMin	397	2	0	1	8	2	17	44	3	13	20	11	1
	PrecStd	170	8	22	19	29	44	46	73	64	77	29	35	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	5.5	10.8	16.1	19.6	22.1	25.6	27.4	24.1	18.9	13.1	6.8
		MCWB	0.3	0.6	4.2	7.5	11.9	16.5	20.3	22.0	19.2	13.3	8.9	4.5
	2%	DB	1.2	3.2	8.1	12.9	17.8	19.3	23.7	25.8	22.5	17.5	11.3	4.6
		MCWB	-1.5	-0.1	3.3	6.7	10.8	15.8	20.3	21.3	18.3	13.3	8.3	2.3
	5%	DB	-0.1	1.8	6.1	10.8	16.0	17.7	22.3	24.5	21.4	16.2	9.9	2.8
		MCWB	-2.6	-1.0	1.5	5.2	10.4	15.3	19.8	21.4	17.6	12.6	7.1	0.3
	10%	DB	-1.4	0.5	4.5	9.3	14.1	16.5	21.3	23.5	20.5	14.9	8.3	1.1
		MCWB	-3.9	-1.9	0.9	4.6	9.9	14.4	19.3	21.0	17.1	11.6	5.5	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.1	5.3	8.9	13.6	17.9	22.1	23.7	21.6	16.1	10.5	5.3
		MCDB	2.3	3.7	9.6	13.5	17.0	19.3	23.8	24.9	23.2	17.0	11.7	6.2
	2%	WB	-0.9	0.8	3.5	7.4	12.4	16.6	21.2	22.9	19.9	14.9	9.2	2.7
		MCDB	0.5	2.2	6.7	10.9	15.6	18.3	22.7	24.3	21.1	16.5	10.6	4.1
	5%	WB	-2.4	-0.5	2.3	6.4	11.4	15.7	20.4	22.2	18.9	13.7	7.5	0.6
		MCDB	-0.3	1.3	5.0	9.5	14.3	17.3	21.7	23.6	20.3	15.3	9.4	2.5
	10%	WB	-3.8	-1.8	1.4	5.5	10.5	14.6	19.7	21.5	18.0	12.2	5.8	-1.0
		MCDB	-1.3	0.5	3.7	8.4	13.2	16.2	20.9	23.0	19.8	14.1	8.0	0.9
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.5	6.7	6.5	5.9	4.7	4.4	4.9	7.2	7.6	7.0	7.3
		MCDBR	8.8	7.6	8.7	9.1	9.3	7.1	6.2	6.0	7.3	8.5	7.5	8.0
		MCWBR	7.6	6.3	5.3	5.0	4.6	4.3	3.9	3.8	4.9	6.5	6.6	7.0
	5% WB	MCDBR	8.4	7.3	7.8	7.8	7.5	6.3	5.1	4.9	5.4	6.8	6.7	7.8
		MCWBR	8.0	6.5	5.5	4.9	4.5	4.2	3.7	3.6	4.4	6.3	6.8	7.5
Clear-Sky Solar Irradiance	taub	0.321	0.383	0.464	0.554	0.576	0.551	0.522	0.477	0.421	0.422	0.382	0.340	
	taud	2.225	2.043	1.877	1.735	1.768	1.917	2.068	2.201	2.277	2.119	2.133	2.144	
	Ebn at Noon	792	785	761	724	723	744	763	786	804	735	701	717	
	Edh at Noon	92	132	176	220	218	189	161	137	118	121	102	92	
All-Sky Solar Radiation	RadAvg	2.21	3.16	4.28	4.80	4.99	5.05	4.49	4.60	4.26	3.23	2.19	1.81	
	RadStd	0.14	0.22	0.27	0.43	0.50	0.60	0.53	0.38	0.39	0.24	0.20	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page